Supplementary Figures

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Figure S6. Relationship between heterozygosity and waterway distance from the downstream-most site (UTA).

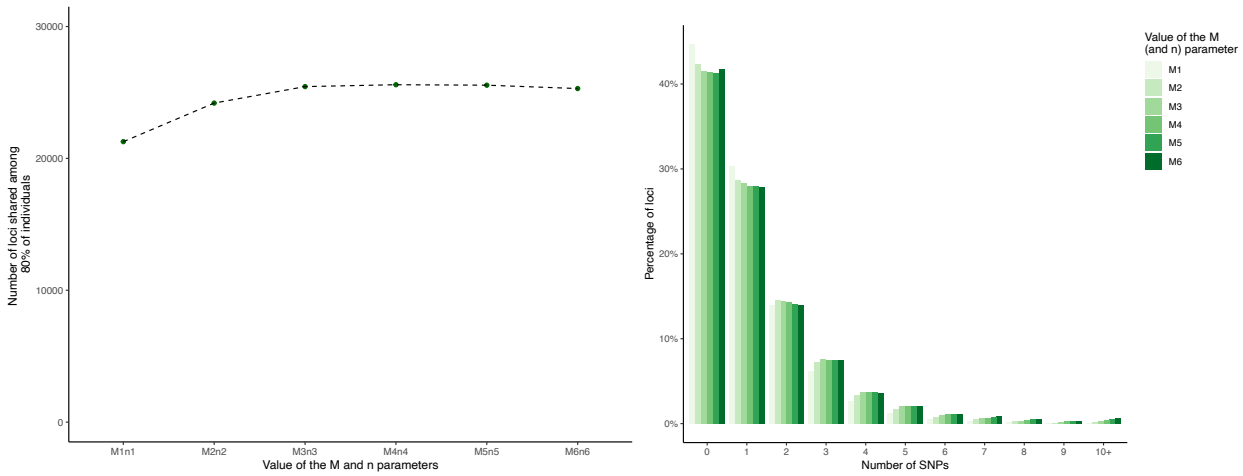

Figure S1. Selection of assembly parameters in the *de novo* analysis using replicate samples. Increasing the *M* and *n* assembly parameters affects the number of polymorphic loci (dashed line) shared by 80% or more of the samples (**A**) and the distribution of the number of Single Nucleotide Ps per locus (**B**). In both plots, the *M* and *n* parameters are kept equal, and the *m* parameter is fixed to 3.

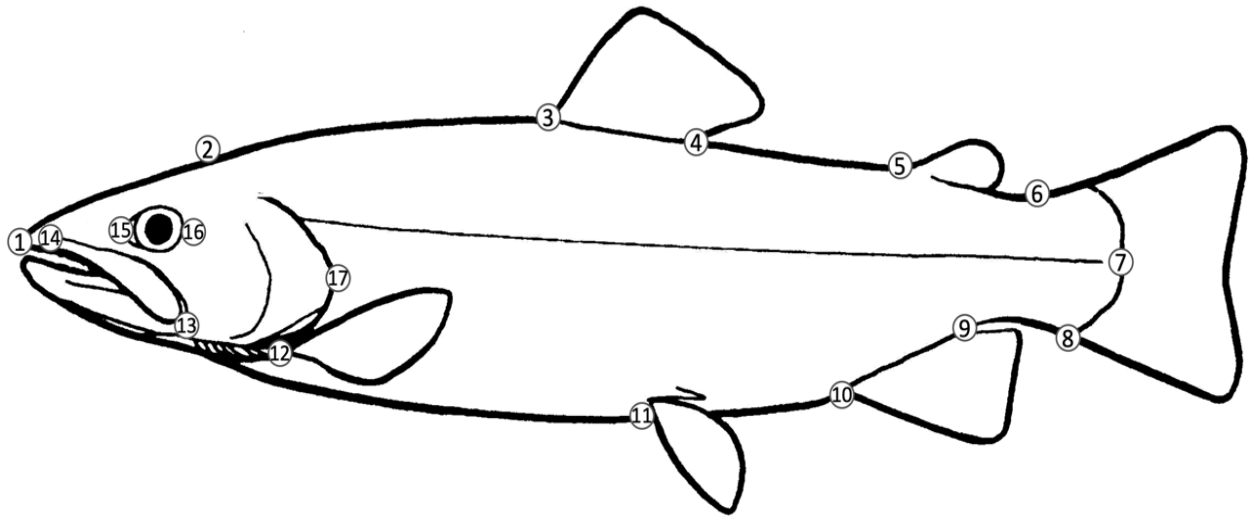

Figure S2 Location of the 17 landmarks used to describe geometric morphometric variation in brook trout body shape: (1) anterior tip of snout, (2) posterior aspect of neurocranium, (3) origin of dorsal fin (4) insertion of dorsal fin, (5) origin of adipose fin, (6) anterior attachment of dorsal membrane from caudal fin, (7) base of middle caudal rays, (8) anterior attachment of ventral membrane from caudal fin, (9) insertion of anal fin, (10) origin of anal fin, (11) origin of pelvic fin, (12) origin of pectoral fin, (13) posterior end of maxillary, (14) anterior end of maxillary, (15) anterior edge of eye, (16) posterior edge of eye, (17) posterior edge of operculum.

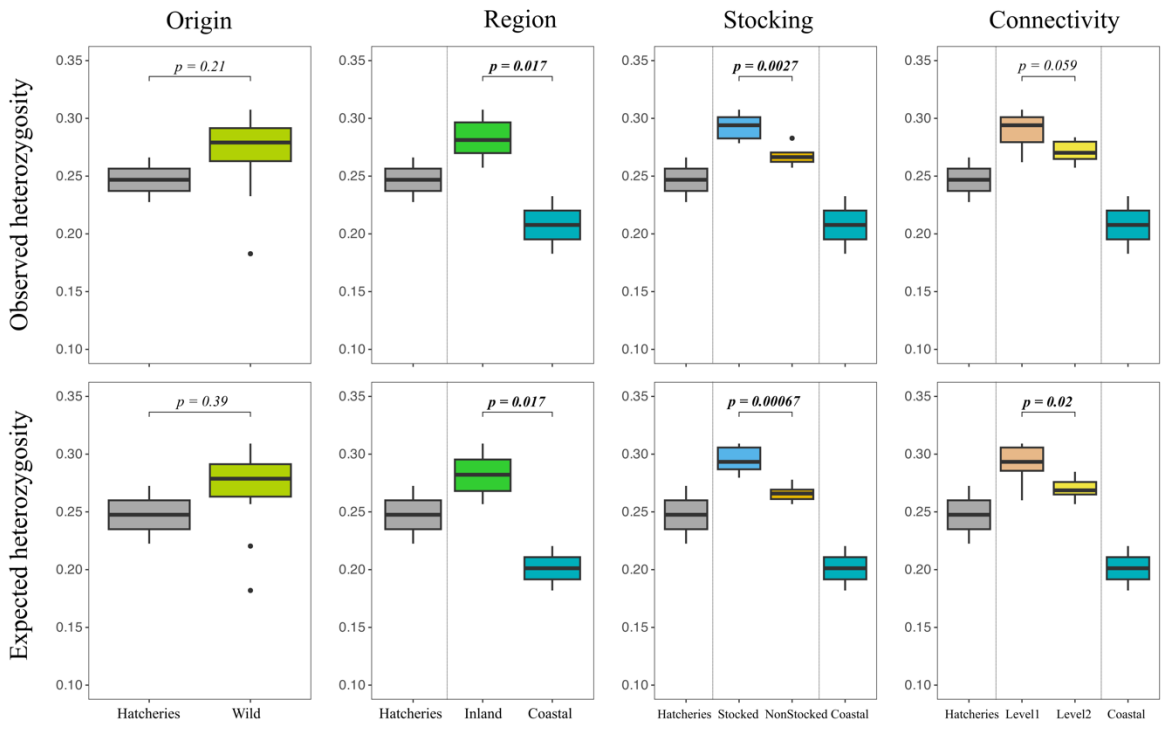

Figure S3. Observed (top row) and expected (bottom row) heterozygosity across four comparisons. Origin of all 16 sites (Wild vs. Hatchery strains). Region of the wild sites (Inland vs. coastal sites). Stocking history (stocked vs. non-stocked sites), coastal sites and hatchery strains as reference. Geographic connectivity among sites; Level1= Below dam, Level2= Above dam. Coastal sites and hatchery strains were used for reference. Significance p-values from Wilcoxon Rank sum exact tests are shown above boxplots of the pairwise comparisons.

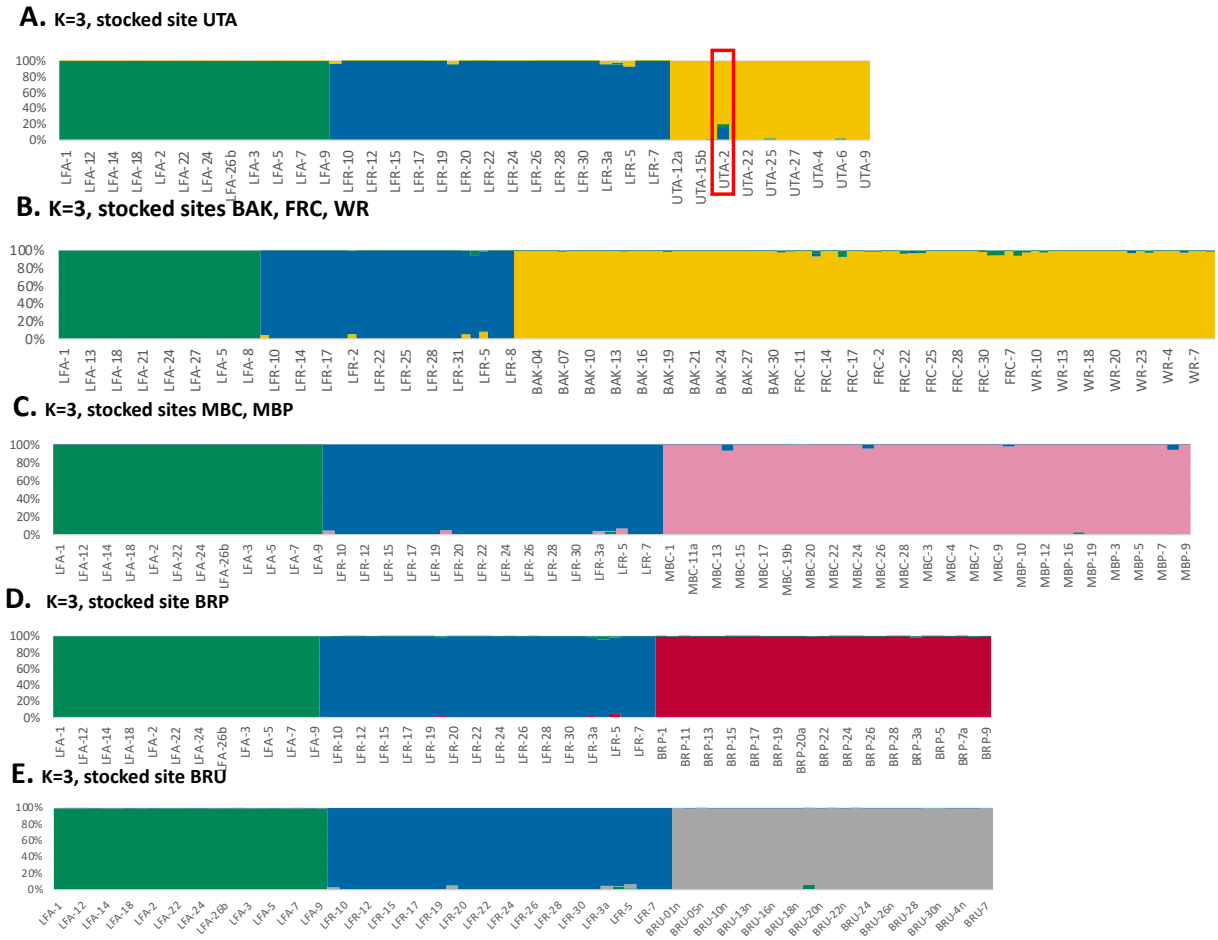

Figure S4. fastStructure analysis of stocked sites with the two hatchery strains. All analyses were run at K=3. Only a single individual (panel-A), UTA-2 (highlighted in red rectangle) had significant (17%) proportion of membership in a hatchery cluster (LFR).

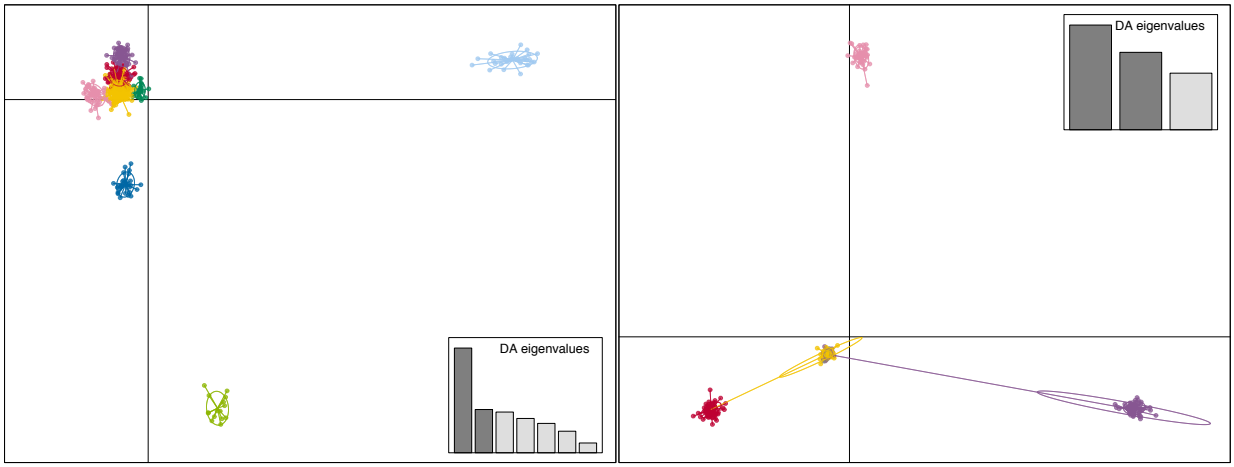

Figure S5. A) DAPC plot of all wild streams and the two hatchery strains. The *find.clusters* algorithm supported K=8 groups applying the lowest BIC criterion. B) DAPC plot of inland sites at K=4 clusters. Sites are color-coded as in the fastStructure plots.

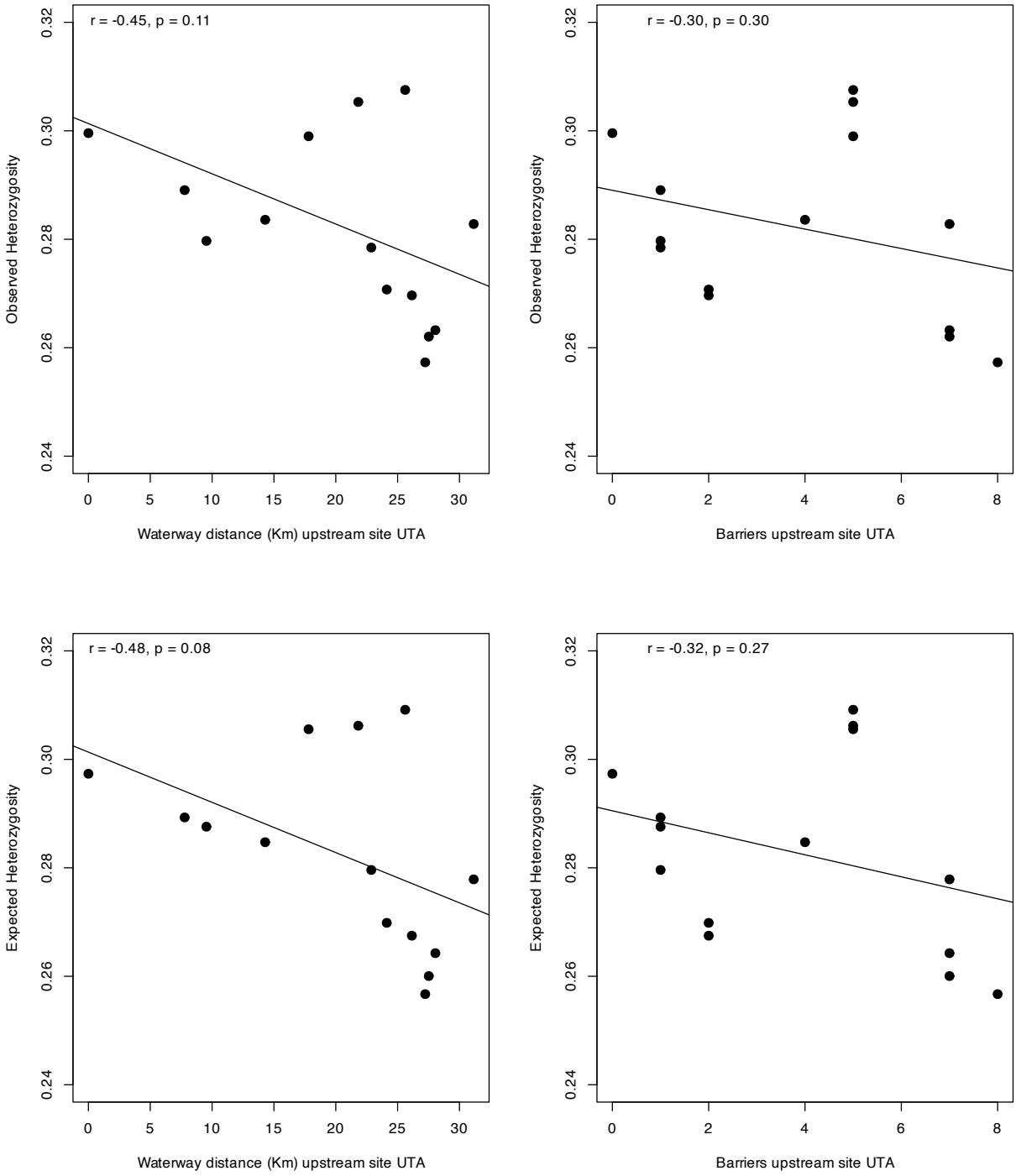

Figure S6. Relationship between genetic diversity, waterway distance and number of barriers from the downstream most site (UTA). Decreased observed (top) and expected (bottom) heterozygosity with increasing distance and number of barriers (dams) from the downstream-most site at UTA, albeit non-significant (Pearson's correlation indicated in each plot).